



KPROJECTXX PRESENTS

HACKFUSION

HARDWARE & EMBEDDED DOMAIN PROBLEM STATEMENTS

ROBOTICS & ROVER NAVIGATION

- **ROB-05 — Autonomous Search & Rescue Rover**

Develop a rover capable of navigating an unknown or partially known environment and detecting potential human presence or hazards. Possible sensors: Camera, LiDAR/ultrasonic, Temperature, Gas sensor.

- **GRI-05 — Autonomous Farm Monitoring Rover**

Develop a small autonomous rover capable of navigating a defined farm area and collecting environmental data. The rover could measure: Soil moisture, Temperature, Humidity, GPS position, and generate a map of the farm conditions.

- **ROB-04 — Autonomous Indoor Delivery Robot**

Develop a small autonomous robot capable of transporting an object between predefined locations while avoiding obstacles. The system should demonstrate: Navigation -> Obstacle avoidance -> Destination identification -> Delivery.

EV & BATTERY TECHNOLOGY

- **PS-EV01: 100,000 mAh Laptop & Router Power Backup**

Design and develop a high-capacity portable UPS/power station capable of providing backup power to devices such as Wi-Fi routers, laptops, CCTV systems and other low-to-medium power electronics.

- Minimum requirements: Laptop DC/USB-C PD output, 12V output for routers, USB output, BMS/protection (over-voltage, over-current, short-circuit), battery level monitoring, charging system, thermal monitoring, portable enclosure.
- Advanced features: Automatic mains failure detection, automatic UPS switchover, estimated remaining runtime, mobile/web monitoring, solar charging, load prioritization.
- Expected output: Working prototype/simulation + circuit architecture + BOM + estimated runtime + efficiency calculations.

SMART ENVIRONMENT & EMBEDDED SYSTEMS

- **PS-ENV01: Design a Low-Cost Smart Air Purification System**

Design and develop a compact, affordable air purifier capable of reducing indoor particulate pollution while continuously monitoring air quality. Combines air filtration with electronic monitoring and intelligent fan control.

- Minimum requirements: Replaceable particulate filter, fan/blower system, air intake and filtered-air outlet, PM2.5 monitoring, temperature monitoring, humidity monitoring, fan-speed control, air-quality indication.
- Advanced features: Automatic fan-speed adjustment based on PM2.5, CO₂ monitoring, VOC monitoring, mobile application/dashboard, filter-life estimation, power consumption monitoring, automatic air-quality history.
- Expected output: Working prototype + airflow design + filter specification + electronics + enclosure concept.

- **EMB-04 — Portable Environmental Monitoring Device**

Build a portable device capable of monitoring multiple environmental parameters such as: Temperature, Humidity, Air quality, CO₂, PM2.5, Atmospheric pressure.

The system should display measurements locally and/or transmit them to a mobile/web dashboard.